Jiaming Yang



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Education

2025.08 – 2028.06 **Harbin Institute of Technology (C9; admitted by recommendation)**
Control Science and Engineering (Master's)

2020.09 – 2024.06 **Harbin Institute of Technology (Project 985, C9) GPA: 3.84 / 4.0**
Automation (Bachelor's)

Internship Experience

2026.06 – 2026.09 **Shanghai SERES Phoenix Technology Co., Ltd. – Data Department**
Autonomous Driving Data Algorithm Engineer

Project background: Contributed to the multi-sensor pre-annotation module of the SACP intelligent-driving data cloud platform. Built an automated data production pipeline for camera and LiDAR data, covering cloud retrieval, detection and tracking, post-processing, quality inspection, and cloud write-back, to provide dynamic and static annotations for manual refinement and downstream perception model training.

Responsibilities:

- **2D quality inspection for dynamic annotations:** Integrated camera 2D detection algorithms into the LiDAR 3D dynamic annotation pipeline to pre-screen anomalies before manual quality inspection. Implemented multi-camera projection of 3D boxes, ego-frame ROI filtering, and 2D–3D instance association; used normalized overlap ratios and the Hungarian algorithm for global matching, and routed anomalies into three categories: suspected false positives, occlusion, and out-of-view cases. Designed task-level quality metrics for 3D precision and 2D recall. Matched 64,600 instance pairs across 8 production-data test CLIPs (3,200 frames), achieving Precision and Recall proxy metrics of **75.0%** and **84.5%** for the main dynamic classes (Car, Truck, Bicycle, Pedestrian). Automatically flagged 12,100 suspected false positives and identified 11,800 potential missed detections to focus manual review on anomalous results;
- **Annotation post-processing pipeline:** Encapsulated dynamic trajectory interpolation using CTRV-EKF and RTS smoothing, together with robust aggregation of static objects in world coordinates, and integrated both algorithms into the dynamic annotation pipeline. Converted 189,200 / 189,500 candidate Features in test cases, achieving **99.86%** processing coverage. Developed frame-wise slicing of aggregated static annotations into ego-frame annotations, combining 6DoF coordinate transformations, five-stream point-cloud fusion, and KD-tree/OBB filtering to generate per-frame ground-truth static-object annotations, reducing the manual static-object annotation workload **to 1/44 of the original level**;
- **Cloud production integration:** Integrated post-processing and quality-inspection modules as pluggable components, supporting result download/upload via S3, Meta write-back, Kafka status reporting, and Docker/Argo deployment. Added exception fallbacks, configuration validation, and automated tests. The implemented features have been integrated into the SACP cloud production workflow to process **approximately 59,000 CLIPs**.

Research Experience

2023.09 – 2024.06 **Deep Learning-Based Object Detection in UAV Aerial Images**
Undergraduate Thesis

Research background: Small-object detection and model compression for UAV aerial imagery.

- Built a UAV aerial-image dataset comprising 5 target classes and 3,224 images;
- Improved YOLOv7 by introducing an “EffectiveSE + BatchNorm” attention mechanism and Network Slimming channel pruning, reducing model parameters by 50% and increasing inference speed by 13.4% while maintaining comparable mAP;
- Published the research at the 44th Chinese Control Conference (“Toward Optical Military Targets: Benchmark Dataset and Lightweight Detection Model”). Passed the thesis defense with a final grade of 92.

2026.09 – Present **Joint Reconstruction with Geometrically Consistent 3D Gaussians from Heterogeneous Aerial and Ground Views**
Master’s Research Focus

Research background: Investigating cross-view geometric association and coordinate-system alignment for 3D reconstruction from aerial and street-view images, providing camera poses and sparse geometry for subsequent joint 3D Gaussian Splatting modeling. The current focus is SfM submodel reconstruction and alignment.

- **Cross-view alignment:** Built a five-stage “retrieval–matching–association–alignment–evaluation” pipeline. Integrated MASt3R for retrieval and matching, associated 2D correspondences with SfM 3D points, and combined RANSAC initialization with seven-parameter Sim(3) optimization constrained by bidirectional reprojection to align aerial and ground coordinate systems. Evaluated the camera poses corresponding to 5,419 ground-view images against a reference in the Park scene of the Horizon-GS dataset, achieving a normalized positional RMSE of **0.286%** of the reference scale and a rotational RMSE of **0.197°**.
- **3D Gaussian reconstruction:** Reproduced 3DGS and 2DGS on self-captured scenes and developed an understanding of the CUDA implementation of differentiable Gaussian rasterization.

Campus Experience

2024.09 – 2025.07 **Kuqa No. 3 Middle School, Aksu Prefecture, Xinjiang Uyghur Autonomous Region**
Grade 10 Physics Teacher

- Received the “Outstanding Young Teacher” and “Outstanding Volunteer Teacher” honors; work featured by China Youth Network and People’s Daily.

2025.09 – Present **School of Astronautics, Harbin Institute of Technology – Automatic Control Theory**
Teaching Assistant

Technical Stack

Programming & Development	C++, Python; Linux, Docker, Git, CMake; PyTorch, OpenCV; Codex, Claude Code
Office Software	Visio, Mermaid, XMind
English	CET-6; fluent spoken English